



FACULTY:

FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

SUBJECT:

WORKSHOP 2

TITLE:

LOGBOOK WORKSHOP 2

PREPARED BY:

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SUPERVISOR:

1) TS. DR. ZURINA BINTI SA'AYA

DATE	DESCRIPTION	NOTES
10 OCTOBER 2023	- We have an online conversation to choose which team will we join. - We discuss what and how many appliances that we need to get from the technician and who will have thumbprint recorded to gain access to the lab(SKK)	
12 OCTOBER 2023	We collect all the appliances after technician approved our form that have been sign by our supervisor	
13 OCTOBER 2023	- First meeting with our supervisors Dr. S.M. Warusia S.M.M. Yassin at his office - Discuss what is our services and the proposal	We create a new group chat using WhatsApp and add our supervisors
16 OCTOBER 2023	- I create a google docs for our proposal. - We distribute all the section of the proposal where each person will be doing 1 or 2 section. - We have a discussion with our second supervisor Ts. Dr. Mohd Fairuz Iskandar Othman	Proposal link : https://docs.google.com/document/d/1bfhxPDUW975D-A1o2gD-KRDpk2LHJHMOVWVWXYq2vmQ/edit?usp=sharing
17 – 20 OCTOBER 2023	We write our proposal and submit it in the ulearn	
23 - 30 OCTOBER 2023	PC , switch and ASA configuration	

2-9
NOVEMBER
2023

Suricata installation and configuration

- I install suricata in our IDS server and made some configuration .
sudo apt install suricata.

```
group6@group6:~$ sudo apt install suricata
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  firefox libevent-pthreads-2.1-7 libhiredis0.14 libhttp2 libhyperscan5 liblua5.1-2
  liblua5.1-common libnet1 libnetfilter-log1 libnetfilter-queue1 oinkmaster snort-rules-def
  suricata-update
Suggested packages:
  snort | snort-pgsql | snort-mysql libtcmalloc-minimal4
The following NEW packages will be installed:
  libevent-pthreads-2.1-7 libhiredis0.14 libhttp2 libhyperscan5 liblua5.1-2 liblua5.1-co
  libnet1 libnetfilter-log1 libnetfilter-queue1 oinkmaster snort-rules-default suricata
  suricata-update
The following packages will be upgraded:
  firefox
1 upgraded, 13 newly installed, 0 to remove and 0 not upgraded.
```

- To enable the suricata.service :
sudo systemctl enable suricata.service
- I stop Suricata each time before editing and testing configuration files:
sudo systemctl stop suricata.service

```
Nov 07 21:47:20 group6 systemd[1]: Started suricata.service - Suricata IDS/IDP daemon.
group6@group6:~$ sudo systemctl stop suricata
group6@group6:~$ sudo systemctl status suricata
* suricata.service - Suricata IDS/IDP daemon
   Loaded: loaded (/lib/systemd/system/suricata.service; enabled; preset: enabled)
   Active: failed (Result: exit-code) since Tue 2023-11-07 21:49:02 +08; 11s ago
     Duration: 49.950s
    Docs: man:suricata(8)
          man:suricatasc(8)
          https://suricata.io/documentation/
```

- There is some error during the installation and configuration that I need to reinstall the package again .

- Command for uninstall suricata:

**sudo apt-get remove
--auto-remove
suricata**

**sudo apt-get purge
suricata**

**sudo apt-get purge
--auto-remove
suricata**

- Some files is missing so I need to reinstall suricata again

- I opened suricata.yaml to edit the configuration:

sudo vim /etc/suricata/suricata.yaml

Look for the bold keyword (ctrl+w) and follow the setting

- **HOME_NET**: [192.168.0.0/23,192.168.2.0/28,192.168.2.24/29.....]
- **community-id**: true
- **af-packet**:
 - **interface**: enp4s0
- **pcap**:
 - **interface**: enp4s0

```
##
## Step 1: Inform Suricata about your network
##

vars:
  # more specific is better for alert accuracy and performance
  address-groups:
    HOME_NET: "[192.168.0.0/23,192.168.2.0/28,192.168.2.16/29,192.168.2.24/29,192.168.2.32/30,]"
    #HOME_NET: "[192.168.0.0/16]"
    #HOME_NET: "[10.0.0.0/8]"
    #HOME_NET: "[172.16.0.0/12]"
    #HOME_NET: "any"

    EXTERNAL_NET: "!$HOME_NET"
    #EXTERNAL_NET: "any"
```

```
pcap:
  - interface: enp4s0
```

```
# Linux high speed capture support
af-packet:
  - interface: enp4s0
```

- To save the configuration using
-sudo vim:
ESC >:wq
-sudo nano:
Ctrl+x > y

- I've run this command to download an up to date ruleset for Suricata:
sudo suricata-update.

- To view list the default set of rule from providers, like ET Open:
sudo suricata-update list-sources

Some of the sources:

```
Name: tgreen/hunting
Vendor: tgreen
Summary: Threat hunting rules
License: GPLv3
Name: malsilo/win-malware
Vendor: malsilo
Summary: Commodity malware rules
License: MIT
Name: stamus/lateral
Vendor: Stamus Networks
Summary: Lateral movement rules
License: GPL-3.0-only
```

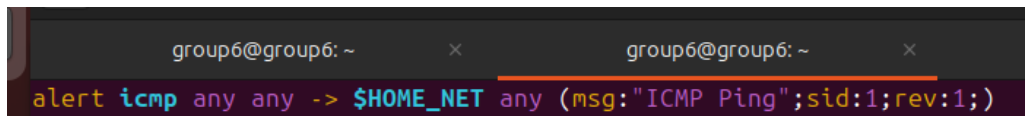
- I've add enable a source from malsilo/win-malware using command :
sudo suricata-udate enable-source malsilo/win-malware

```
04:43:43 - <Info> -- Saved /var/lib/suricata/update/cache/index.yaml
:-$ sudo suricata-update enable-source malsilo/win-malware
04:43:53 - <Info> -- Using data-directory /var/lib/suricata.
04:43:53 - <Info> -- Using Suricata configuration /etc/suricata/suricata.yaml
04:43:53 - <Info> -- Using /etc/suricata/rules for Suricata provided rules.
04:43:53 - <Info> -- Found Suricata version 7.0.0 at /usr/bin/suricata.
04:43:53 - <Warning> -- The source malsilo/win-malware is already enabled.
04:43:53 - <Info> -- Source malsilo/win-malware enabled
```

- I've also add new rule at local.rules file.The command is :

sudo vim /etc/suricata/rules/local.rules

The rules is :



```
group6@group6: ~
alert icmp any any -> $HOME_NET any (msg:"ICMP Ping";sid:1;rev:1;)
```

Action: alert

Protocol: icmp

Source ip: any

Source port : any

Destination ip: \$HOME_NET

Destination port: any

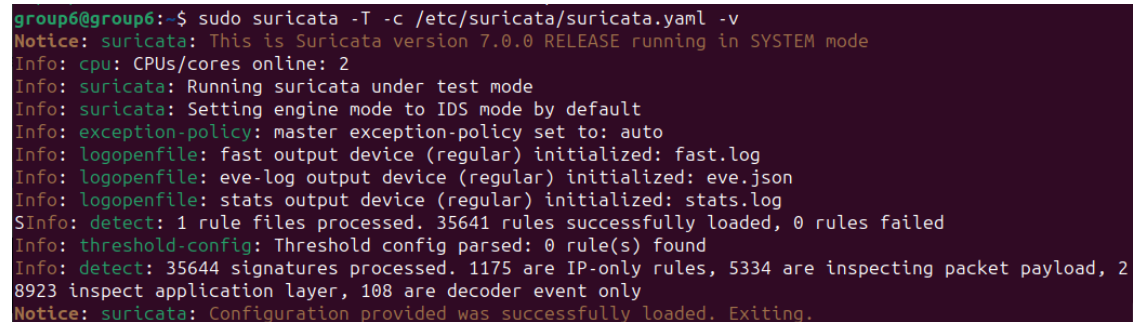
Message: ICMP Ping

Sid (unique signature number): 1

Rev (revision number): 1

- I've use this command to check if there are any error when writing the rules in local.rules :

sudo suricata -T -c /etc/suricata/suricata.yaml -v



```
group6@group6:~$ sudo suricata -T -c /etc/suricata/suricata.yaml -v
Notice: suricata: This is Suricata version 7.0.0 RELEASE running in SYSTEM mode
Info: cpu: CPUs/cores online: 2
Info: suricata: Running suricata under test mode
Info: suricata: Setting engine mode to IDS mode by default
Info: exception-policy: master exception-policy set to: auto
Info: logopenfile: fast output device (regular) initialized: fast.log
Info: logopenfile: eve-log output device (regular) initialized: eve.json
Info: logopenfile: stats output device (regular) initialized: stats.log
SInfo: detect: 1 rule files processed. 35641 rules successfully loaded, 0 rules failed
Info: threshold-config: Threshold config parsed: 0 rule(s) found
Info: detect: 35644 signatures processed. 1175 are IP-only rules, 5334 are inspecting packet payload, 2
8923 inspect application layer, 108 are decoder event only
Notice: suricata: Configuration provided was successfully loaded. Exiting.
```

- The rule is added to alert any ICMP ping packets from any source IP and port to any destination IP and port within the \$HOME_NET variable and display message -"ICMP Ping"

- To start the Suricata and check suricata status , I use :

sudo systemctl start suricata.service

sudo systemctl status suricata.service

```
group6@group6:~$ sudo systemctl status suricata
● suricata.service - Suricata IDS/IDP daemon
   Loaded: loaded (/lib/systemd/system/suricata.service; enabled; preset: enabled)
   Active: active (running) since Tue 2023-11-07 22:49:04 +08; 10s ago
     Docs: man:suricata(8)
           man:suricatasc(8)
           https://suricata.io/documentation/
  Process: 4077 ExecStart=/usr/bin/suricata -D --af-packet -c /etc/suricata/suricat
 Main PID: 4078 (Suricata-Main)
    Tasks: 1 (limit: 2175)
   Memory: 278.9M
      CPU: 10.188s
   CGroup: /system.slice/suricata.service
```

- To test this rule, I send a ping command from other machine using cmd to this machine.

ping 192.168.2.18

- Then in the machine terminal, I use command this command to view the latest output in the fast.log:

sudo tail -f /var/log/suricata/fast.log

```
group6@group6:~$ sudo tail -f /var/log/suricata/fast.log
[sudo] password for group6:
11/08/2023-10:38:26.964133  [**] [1:1:1] ICMP Ping [**] [Classification: (null)] [Priority: 3] {ICMP} 192.168.2.2:8 -> 192.168.2.18:0
11/08/2023-10:38:26.964201  [**] [1:1:1] ICMP Ping [**] [Classification: (null)] [Priority: 3] {ICMP} 192.168.2.2:8 -> 192.168.2.18:0
11/08/2023-10:38:26.964453  [**] [1:1:1] ICMP Ping [**] [Classification: (null)] [Priority: 3] {ICMP} 192.168.2.18:0 -> 192.168.2.2:0
11/08/2023-10:38:26.964578  [**] [1:1:1] ICMP Ping [**] [Classification: (null)] [Priority: 3] {ICMP} 192.168.2.18:0 -> 192.168.2.2:0
```

11-15 NOVEMBER 2023	<u>Port Mirroring</u> - I've configured port mirroring using Putty by connecting my laptop to the switch using console wire. - These are the step to configure port mirroring: Step 1: Enter the configuration mode: configure - Step 2: Enable the port mirror function and set the destination port monitor session 1 destination interface gigabitEthernet 1/0/2 - Step 3: Configure ports as the monitored interfaces monitor session 1 source interface gigabitEthernet 1/0/1 both monitor session 1 source interface gigabitEthernet 1/0/3 both monitor session 1 source interface gigabitEthernet 1/0/4 both monitor session 1 source interface gigabitEthernet 1/0/5 both - Step 4: Return to privileged EXEC mode. end - Step 5: Verify the Port Mirror configuration. show monitor session 1	“Both” at the end of sentence in the step 3 means the incoming(Ingress) and outgoing(Egress) packets on source port can be copied to the destination port. - To revert the command, I use: no monitor session 1 destination interface gigabitEthernet 1/0/2
24 NOVEMBER 2023	First progress update with TS. DR. ZURINA BINTI SA'AYA	

11 DECEMBER 2023	First progress update with Dr Fairuz	Some of the comments given by Dr Fairuz: - Need to add more rules. - Need to understand the alert . - Use trunk as source port for mirroring.
11 - 16 DECEMBER 2023	To visualize the suricata logs , I used Elastic Stack including Filebeat <u>Install and configure Elasticsearch:</u> Before installing Elasticsearch and Kibana, I've added a Elastic GPG key to my server and create a repository for the Elasticsearch package by creating the file /etc/apt/sources.list.d/elastic-7.x.list using below commands: <pre>curl -fsSL https://artifacts.elastic.co/GPG-KEY-elasticsearch sudo gpg --dearmor -o /usr/share/keyrings/elastic.gpg</pre> <pre>echo "deb [signed-by=/usr/share/keyrings/elastic.gpg] https://artifacts.elastic.co/packages/7.x/apt stable main" sudo tee -a /etc/apt/sources.list.d/elastic-7.x.list</pre> <pre>group6@group6:~\$ curl -fsSL https://artifacts.elastic.co/GPG-KEY-elasticsearch sudo gpg --dearmor -o /usr/share/keyrings/elastic.gpg group6@group6:~\$ echo "deb [signed-by=/usr/share/keyrings/elastic.gpg] https://artifacts.elastic.co/packages/7.x/apt stable main" sudo tee -a /etc/apt/sources.list.d/elastic-7.x.list</pre>	During this week, I've tried to install the software(Elasticsearch ,kibana,filebeat) many times because there are errors when I tried to start elasticsearch.service. After a while. I've successfully install Elasticsearch.

- The first step is to install elasticsearch:
sudo apt install elasticsearch

```
group6@group6:~$ sudo apt install elasticsearch
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  elasticsearch
0 upgraded, 1 newly installed, 0 to remove and 1 not upgraded.
Need to get 323 MB of archives.
After this operation, 540 MB of additional disk space will be used.
Get:1 https://artifacts.elastic.co/packages/7.x/apt/stable/main amd64 elasticsearch amd64 7.10.2 [323 MB]
```

- I've edit the configuration in elasticsearch.yml using command:
sudo nano /etc/elasticsearch/elasticsearch.yml.
- Elasticsearch only accepts local connections by default. I change it so that Kibana can access it over the private IP address.

```
# ----- Network -----
#
# By default Elasticsearch is only accessible on localhost. Set a different
# address here to expose this node on the network:
#
#network.host: 192.168.0.1
network.bind_host: ["127.0.0.1","192.168.2.18"]
# By default Elasticsearch listens for HTTP traffic on the first free port it
# finds starting at 9200. Set a specific HTTP port here:
#
```

- Command to uninstall Elasticsearch:

sudo systemctl stop elasticsearch.service

sudo apt-get purge elasticsearch

sudo rm -rf /etc/elasticsearch

sudo rm -rf /usr/share/elasticsearch

sudo rm -rf /var/lib/elasticsearch

sudo rm /etc/apt/sources.list.d /elastic-7.x.list -

- Add this lines at the last line of the file

```
# https://www.elastic.co/guide/en/elasticsearch/reference/7.10/_config.html
discovery.type: single-node
xpack.security.enabled: true
```

- This is to enabling the security features and ensure that Elastic is configured to run on a single node
- I then add the proper firewall rules for Elasticsearch so that it is accessible via the private network.

```
group6@group6:~$ sudo ufw allow in on enp4s0
Rule added
Rule added (v6)
group6@group6:~$ sudo ufw allow out on enp4s0
Rule added
Rule added (v6)
```

- I then reload the service daemon, enable the elasticsearch and start the service using command below:

```
sudo systemctl daemon-reload
sudo systemctl enable elasticsearch
sudo systemctl start elasticsearch
```

- I check firewall status by running command:
sudo ufw status

- Then I run this command to view elasticsearch status:
sudo systemctl status elasticsearch

```
group6@group6:~$ sudo systemctl status elasticsearch
● elasticsearch.service - Elasticsearch
   Loaded: loaded (/lib/systemd/system/elasticsearch.service; enabled; preset: enabled)
   Active: active (running) since Mon 2023-12-11 21:58:41 +08; 2min 22s ago
     Docs: https://www.elastic.co
   Main PID: 31385 (java)
    Tasks: 68 (limit: 2175)
   Memory: 982.8M
      CPU: 1min 14.335s
   CGroup: /system.slice/elasticsearch.service
           └─31385 /usr/share/elasticsearch/jdk/bin/java -Xms4m -Xmx64m -XX:+UseSerialGC -Dcli
```

- To get a generated random password by elasticsearch , I've use a command :

cd /usr/share/elasticsearch/bin
sudo ./elasticsearch-setup-passwords auto

- I copied the output and save it in the desktop as passwordauto

```
group6@group6:~/Desktop$ sudo cat passwordauto
Changed password for user apm_system
PASSWORD apm_system = nTOJx8BR680KWIfuitH0

Changed password for user kibana_system
PASSWORD kibana_system = GNO20b1maHJLzIRbbjG1

Changed password for user kibana
PASSWORD kibana = GNO20b1maHJLzIRbbjG1

Changed password for user logstash_system
PASSWORD logstash_system = KfADa4qPUmi5cfFv2Bg5

Changed password for user beats_system
PASSWORD beats_system = wE6HVwBLxNNXUi3LTes4

Changed password for user remote_monitoring_user
PASSWORD remote_monitoring_user = 02V9wbuxoyrwm7Y1amxw

Changed password for user elastic
PASSWORD elastic = rPOBAAKrcnKmt4f6vaAW
```

Install and configure Kibana:

- I use this command to install kibana :
sudo apt install kibana

```
group6@group6:~$ sudo apt install kibana
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  kibana
0 upgraded, 1 newly installed, 0 to remove and 6 not upgraded.
Need to get 298 MB of archives.
After this operation, 770 MB of additional disk space will be used.
Get:1 https://artifacts.elastic.co/packages/7.x/apt/stable/main amd64 kibana amd64 7.17.15 [298 MB]
Fetched 298 MB in 22s (13.7 MB/s)
Selecting previously unselected package kibana.
Reading database ... 163711 files and directories currently installed.)
Preparing to unpack .../kibana_7.17.15_amd64.deb ...
Unpacking kibana (7.17.15) ...
Setting up kibana (7.17.15) ...
Creating kibana group... OK
Creating kibana user... OK
```

- The first thing I do is enabling the xpack security function by generating secret keys. Kibana uses these secret keys to store data in Elasticsearch. The utility to generate secret keys can be accessed from the /usr/share/kibana/bin directory:
cd /usr/share/kibana/bin/
sudo ./kibana-encryption-keys generate -q --force

```
xpack.encryptedSavedObjects.encryptionKey: af8295e92163bf3d45dd983cbb681f1
xpack.reporting.encryptionKey: 51c4ba47e2bc5a655c928ad173fba9c7
xpack.security.encryptionKey: e5ecef7a4827ae8344098a161523e7ee
```

- I copied the output above and paste at the end file kibana.yml.x. Kibana needs to be configured so that it's accessible on the server's private IP address. I find the line where it has #server.host: "localhost" in the file and change it to machine IP address

```
# Kibana is served by a back end server. This setting specifies the port to use.
#server.port: 5601

# Specifies the address to which the Kibana server will bind. IP addresses and h
# The default is 'localhost', which usually means remote machines will not be ab
# To allow connections from remote users, set this parameter to a non-loopback a
server.host: "192.168.2.18"
```

- I run this command to set username and password for authentication.:
cd /usr/share/kibana/bin
sudo ./kibana-keystore add elasticsearch.username - kibana_system
sudo ./kibana-keystore add elasticsearch.password - (look at passwordauto)
- Then I start and enable the service:
sudo systemctl enable kibana --now
sudo systemctl status kibana

```
group6@group6:~$ sudo systemctl status kibana
● kibana.service - Kibana
   Loaded: loaded (/etc/systemd/system/kibana.service; enabled; preset: enabled)
   Active: active (running) since Mon 2023-12-11 21:58:04 +08; 6s ago
     Docs: https://www.elastic.co
   Main PID: 31475 (node)
    Tasks: 11 (limit: 2175)
   Memory: 166.7M
      CPU: 4.465s
   CGroup: /system.slice/kibana.service
           └─31475 /usr/share/kibana/bin/./node/bin/node /usr/share/kibana/bin/./src/cli/dist
```

Install and configure Filebeat:

- I use this command to install filebeat :
sudo apt install filebeat
- To connect Kibana's dashboard, open filebeat.yml using command :
sudo nano /etc/filebeat/filebeat.yml
- I find the line #host: "localhost:5601" in the Kibana section and add the following line right below it as shown.

```
setup.kibana:
```

```
# Kibana Host
# Scheme and port can be left out and will be set to the default (http and 5601)
# In case you specify an additional path, the scheme is required: http://localhost:5601/path
# IPv6 addresses should always be defined as: https://[2001:db8::1]:5601
#host: "localhost:5601"
host: "192.168.2.18:5601"
# Kibana Space ID
```

- Next, I find the Elasticsearch Output section of the file and edit the values of hosts, username, and password. For the username I put "elastic" and for the password I referred the password for elastic system in passwordauto file

- Command to uninstall Filebeat:

sudo systemctl stop filebeat

sudo apt-get purge filebeat

sudo rm -r /var/lib/filebeat

sudo rm -r /etc/filebeat

	<pre>output.elasticsearch: # Array of hosts to connect to. hosts: ["192.168.2.18:9200"] # Protocol - either 'http' (default) or 'https'. #protocol: "https" # Authentication credentials - either API key or username/password. #api_key: "id:api_key" username: "elastic" password: "rPOBAAKrcnKMT4f6vaAW"</pre> • Next, I've enable Filebeat's built-in Suricata module using command : sudo filebeat modules enable suricata • The last step is to load the SIEM dashboards and pipelines into Elasticsearch using command: sudo filebeat setup <pre>group6@group6:~\$ sudo filebeat setup Overwriting ILM policy is disabled. Set `setup.ilm.overwrite: true` for enabling. Index setup finished. Loading dashboards (Kibana must be running and reachable) Loaded dashboards Setting up ML using setup --machine-learning is going to be removed in 8.0.0. Please use the ML app instead. See more: https://www.elastic.co/guide/en/machine-learning/current/index.html It is not possible to load ML jobs into an Elasticsearch 8.0.0 or newer using the Beat. Loaded machine learning job configurations Loaded Ingest pipelines</pre> • Then I started the filebeat service: sudo systemctl start filebeat	• When trying to use this command, I get error that not allowing me to load the SIEM dashboard and pipelines. • To solve this problem , I tried to stop and restart all the services including suricata .Then I try to run the command again . • To stop and restart all the service, I used command : sudo systemctl stop suricata elasticsearch kibana filebeat
--	---	---

	<pre>group6@group6:~\$ sudo systemctl status filebeat ● filebeat.service - Filebeat sends log files to Logstash or directly to Elasticsearch. Loaded: loaded (/lib/systemd/system/filebeat.service; enabled; preset: enabled) Active: active (running) since Mon 2023-12-11 21:56:45 +08; 2s ago Docs: https://www.elastic.co/beats/filebeat Main PID: 31309 (filebeat) Tasks: 7 (limit: 2175) Memory: 133.1M CPU: 553ms CGroup: /system.slice/filebeat.service └─31309 /usr/share/filebeat/bin/filebeat --environment systemd -c /etc/filebeat/filebeat.yml</pre>	sudo systemctl restart suricata elasticsearch kibana filebeat
17-23 DECEMBER 2023	<u>Install and configure SSH LocalTunnel</u> - To install openssh-server: sudo apt install openssh-server - Then I enable and start the sshd service: sudo systemctl enable sshd sudo systemctl start sshd - To create the SSH Tunnel : ssh -L 5601:192.168.2.18:5601group6@192.168.2.18 -N - The -L flag refers to the local SSH Tunnel, which forwards traffic from your PC's port to the server. - The -N flag tells OpenSSH not to execute any command but keep the connection alive as long as the tunnel runs. <pre>^Cgroup6@group6:~\$ ssh -L 5601:192.168.2.18:5601 group6@192.168.2.18 -N group6@192.168.2.18's password:</pre> - Now that the tunnel is open, I accessed Kibana by opening the URL http://localhost:5601 on my PC's browser.	The password is PC's password



Welcome to Elastic

You have logged out of Elastic.

Username

elastic

Password

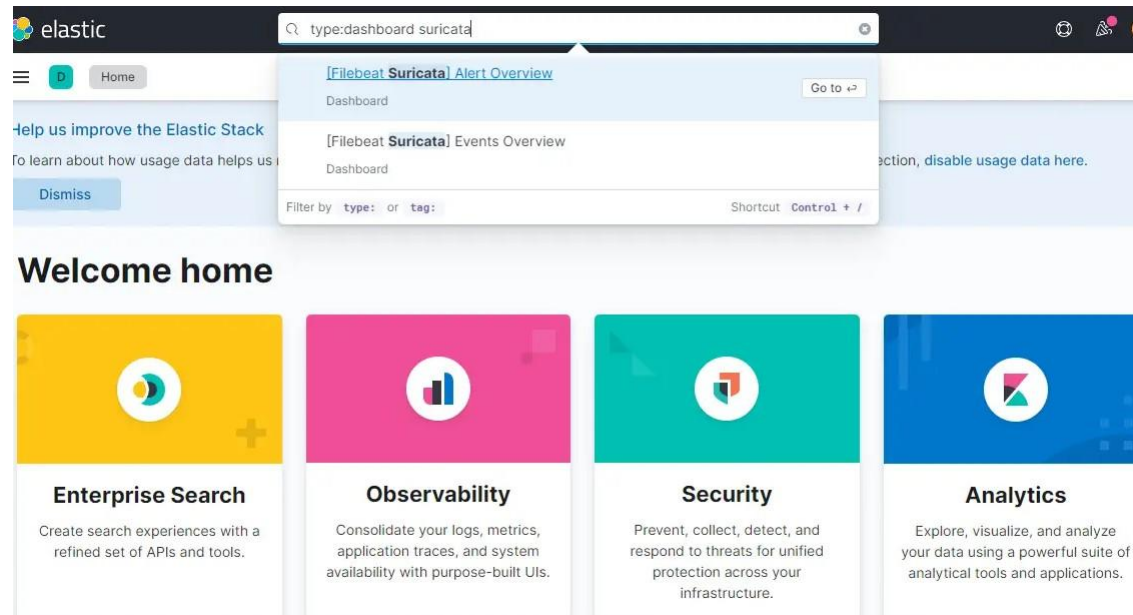
 rPOBAAKrcnKMT4f6vaAW|

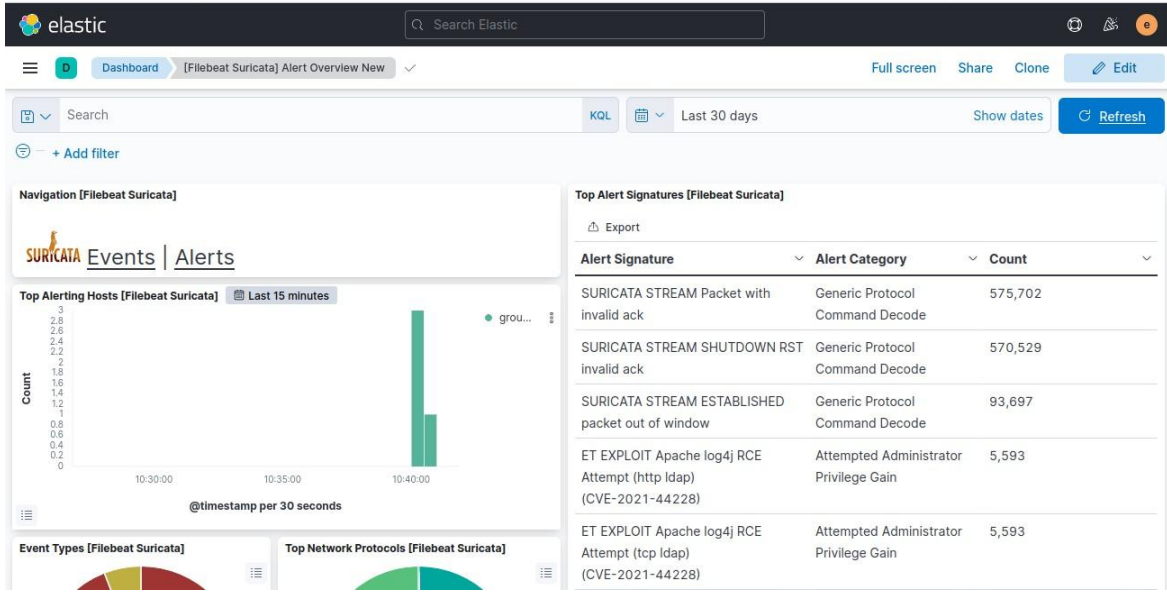


Log in

- Username and password is saved in **/etc/filebeat/filebeat.yml**

- I've searched “**type:data suricata**” in the search box at the top to locate Suricata's information.



	 Top Alert Signatures [Filebeat Suricata] <table border="1"> <thead> <tr> <th>Alert Signature</th> <th>Alert Category</th> <th>Count</th> </tr> </thead> <tbody> <tr> <td>SURICATA STREAM Packet with invalid ack</td> <td>Generic Protocol Command Decode</td> <td>575,702</td> </tr> <tr> <td>SURICATA STREAM SHUTDOWN RST invalid ack</td> <td>Generic Protocol Command Decode</td> <td>570,529</td> </tr> <tr> <td>SURICATA STREAM ESTABLISHED packet out of window</td> <td>Generic Protocol Command Decode</td> <td>93,697</td> </tr> <tr> <td>ET EXPLOIT Apache log4j RCE Attempt (http ldap) (CVE-2021-44228)</td> <td>Attempted Administrator Privilege Gain</td> <td>5,593</td> </tr> <tr> <td>ET EXPLOIT Apache log4j RCE Attempt (tcp ldap) (CVE-2021-44228)</td> <td>Attempted Administrator Privilege Gain</td> <td>5,593</td> </tr> </tbody> </table>	Alert Signature	Alert Category	Count	SURICATA STREAM Packet with invalid ack	Generic Protocol Command Decode	575,702	SURICATA STREAM SHUTDOWN RST invalid ack	Generic Protocol Command Decode	570,529	SURICATA STREAM ESTABLISHED packet out of window	Generic Protocol Command Decode	93,697	ET EXPLOIT Apache log4j RCE Attempt (http ldap) (CVE-2021-44228)	Attempted Administrator Privilege Gain	5,593	ET EXPLOIT Apache log4j RCE Attempt (tcp ldap) (CVE-2021-44228)	Attempted Administrator Privilege Gain	5,593	
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24 - 30 DECEMBER 2023	I've change the port mirroring configuration. Source port: monitor session 1 source interface gigabitEthernet 1/0/4 both Destination port: monitor session 1 destination interface gigabitEthernet 1/0/8	interface gigabitEthernet 1/0/4 is a trunk port																		

GUI view:

Port Mirroring Session List

Session	Destination Port	Mode	Source Interfaces
1	1/0/8	Ingress Only	1/0/4
		Egress Only	
		Both	
Total: 1			

CLI view:

```
monitor session 1 source interface gigabitEthernet 1/0/4 both
monitor session 1 destination interface gigabitEthernet 1/0/8
```

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- I've add a new rules at local.rules file.

```
group6@gr... x group6@gr... x group6@gr... x group6@gr... x group6@gr... x group6@gr... x
alert icmp any any -> $HOME_NET any (msg:"ICMP Ping";sid:1;rev:1;)
alert tcp any any -> $HOME_NET any (msg:"TCP Port Scanning"; flags: S,12; threshold: type both
, track by_src, count 10, seconds 10; classtype:attempted-recon; sid:2; rev:1; metadata:created
_at 2023_12_25, updated_at 2023_12_25;)
alert http any any -> $HOME_NET any (msg:"Possible HTML Injection attack(Contains script tag)";
flow:established,to_server; content:"cookie";nocase; http_uri; sid:9; rev:1;)
```

	Second Rules	Third Rule
Action	alert	alert
Protocol	tcp	http
Src Ip	any	any
Src Port	any	any
Dst Ip	\$HOME_NET	\$HOME_NET
Dst port	any	any
Message	TCP Port Scanning	Possible HTML Injection attack(Contains script tag)
Sid	2	9
Rev	1	1

- Before testing the rule, I run this command to make sure the rules is bening update:

- The second rule will detect possible port scans on every port. It also uses a threshold to limit the number of alerts per destination IP address within a 10-second window.
- The third rule will detects possible HTML injection attacks that contain a script tag. It matches HTTP packets that have the word “cookies” in the URI, regardless of the case. It only applies to established flows from any source to the home network.

sudo suricata-update

sudo suricata -T -c /etc/suricata/suricata.yaml -v

sudo systemctl restart suricata

- I've test the second rule using kali VM and run this command in the terminal:

sudo nmap -sS 192.168.2.2

```
(yan@yan)-[~]
$ sudo nmap -sS 192.168.2.2
Starting Nmap 7.94 ( https://nmap.org ) at 2024-01-10 23:10 +08
Nmap scan report for group6.group6.com (192.168.2.2)
Host is up (0.00090s latency).
Not shown: 982 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
53/tcp    open  domain
80/tcp    open  http
88/tcp    open  kerberos-sec
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
389/tcp   open  ldap
443/tcp   open  https
445/tcp   open  microsoft-ds
464/tcp   open  kpasswd5
593/tcp   open  http-rpc-epmap
636/tcp   open  ldapssl
992/tcp   open  telnet
3268/tcp  open  globalcatLDAP
3269/tcp  open  globalcatLDAPssl
5357/tcp  open  wsdapi
```

- To view only the alert for the second rule, I use this command:

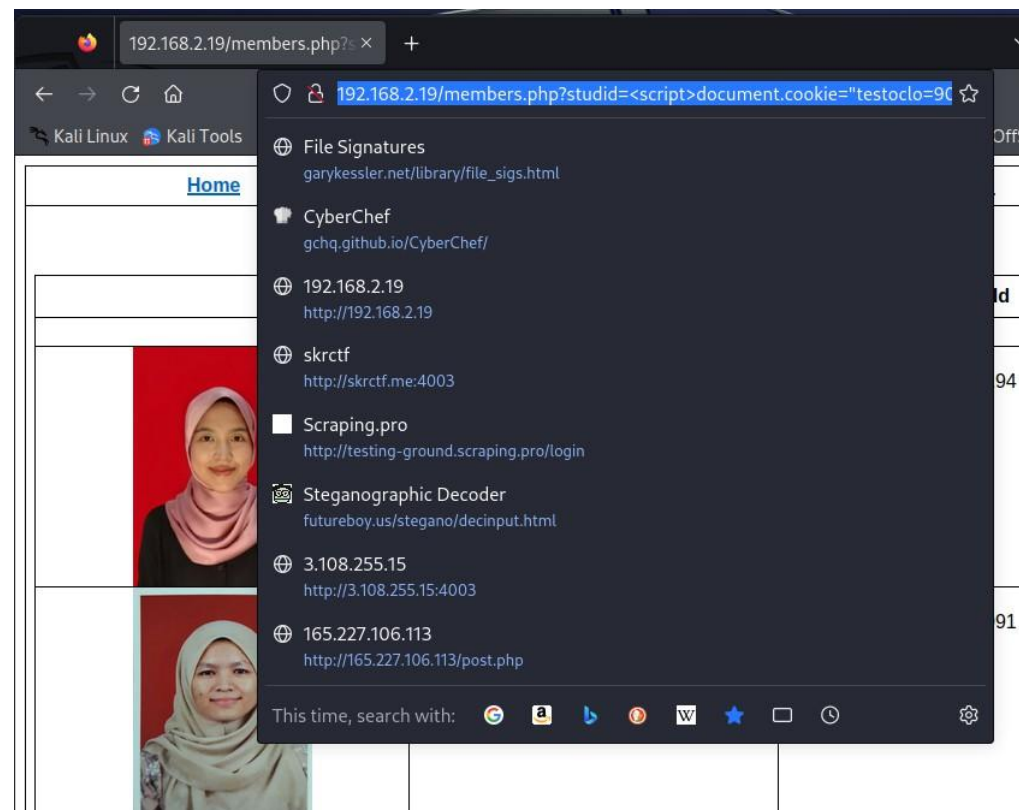
grep -r "TCP Port Scanning" /var/log/suricata/fast.log

```
group6@group6:~$ grep -r "TCP Port Scanning" /var/log/suricata/fast.log
01/09/2024-13:40:15.556833  [**] [1:2:1] TCP Port Scanning [**] [Classification: Attempted Information Leak] [Priority: 2] {TCP}
192.168.0.5:52144 -> 192.168.2.2:53
```

- This is the alert in the elastic dashboard based on the above test

>	Jan 9, 2024 @ 13:48:15.920	group6	192.168.0.5	52144	192.168.2.2	53	-	-
---	----------------------------	--------	-------------	-------	-------------	----	---	---

- I've tested the third rule also using kali VM and open our web server which is 192.168.2.19 using firefox.



- At the member page, I ‘ve added a script that may contain injection attack and reload the page.

The script is:

%3Cscript%3Edocument.cookie=%22testoclo=9022;%22%3C/script%3E

- To view the alert that being generated, I run command:

grep -r “Possible HTML Injection attack” /var/log/suricata/fast.log

```
group6@group6:~$ grep -r "Possible HTML Injection attack" /var/log/suricata/fast.log
01/10/2024-23:11:56.870958  [**] [1:9:1] Possible HTML Injection attack(contains script tag) [**] [Classification: (null)] [Priority: 3] {TCP} 192.168.0.10:50294 -> 192.168.2.19:80
```

- **At Elastic Dashboard**

>	Jan 10, 2024 @ 23:11:56.288	group6	192.168.0.10	34136	192.168.2.19	80
---	-----------------------------	--------	--------------	-------	--------------	----

ricata]		Limited to 50
network.transport	tcp	
related.hosts	192.168.2.19	
related.ip	192.168.0.10, 192.168.2.19	
rule.id	9	
rule.name	Possible HTML Injection attack(Contains script tag)	
service.type	suricata	
source.address	192.168.0.10	
source.bytes	716B	
source.ip	192.168.0.10	
source.packets	4	

15 JANUARY 2024	• Last Demo with Dr Fairuz	
17 JANUARY 2024	• Demo with evaluator	